Four APS follow-up leads: implementation and correctness

28 October 2026

Scope. Native bounded development checks, not a performance result.

Baseline: `6187036c049b6de2b1521b014f575952e4f1705c`. The earlier APS implementation is PR 2614.

1. Verified early termination. The new visitor receives roots only after evaluation against the original equations. Coordinates are decoded and deduplicated before checking actual signed point sums, known labels, $B \neq 0$, and $dG = Q$. Uninformative roots do not stop the search. Hybrid guesses and covering partitions share a cooperative deadline and stop on the verified witness. A stopped search explicitly reports partial enumeration; it is distinct from complete exhaustion and timeout. Exhaustive mode and the legacy full-field solver remain available as controls.

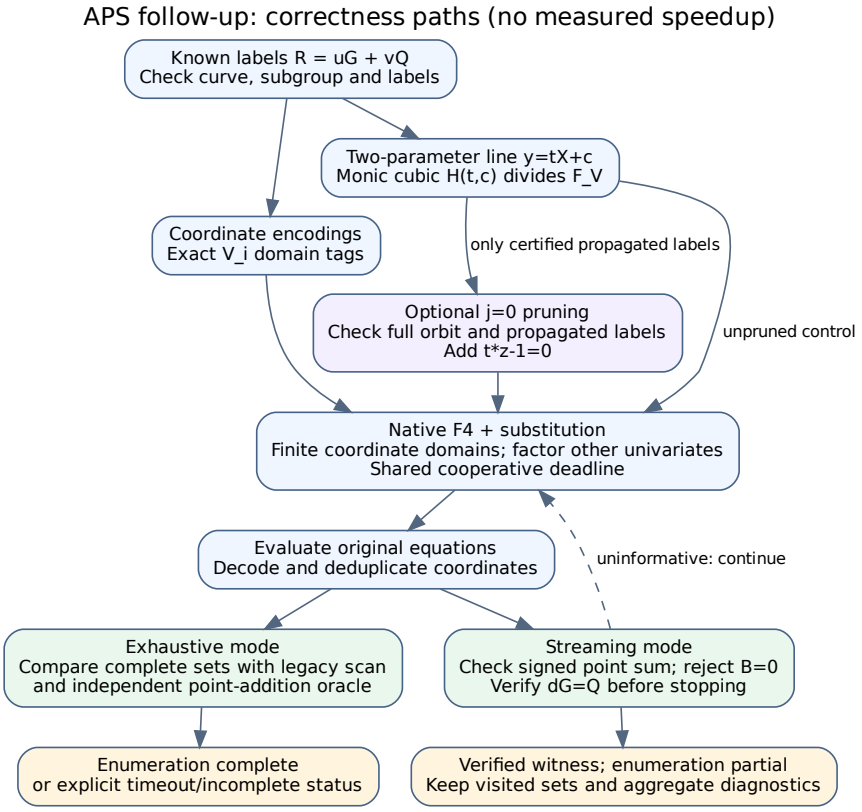
2. Membership-aware root extraction. Coordinate variables carry exact finite domains generated together with their vanishing equations. Substitution preserves the domain's original variable identity. Auxiliary, elementary-symmetric and line parameters remain unrestricted. Their univariates use the existing counted $\gcd(f, X^p - X)$ and Cantor–Zassenhaus implementation with cancellation. If there is no univariate, full-field fallback remains possible and explicit. The field cap stays $p \leq 65521$. Root diagnostics are stage observations; they omit other costs and cannot establish a complete-cost gain.

3. Two-parameter line selection. For $E : y^2 = x^3 + ax + b$ and $y = tx + c$, substitution gives

$$H_{t,c}(X) = X^3 - t^2X^2 + (a - 2tc)X + b - c^2.$$

The three remainder equations for $H_{t,c} \mid F_V$ select distinct base coordinates. This uses two parameters instead of three elementary symmetric coordinates plus the symmetric S_3 equation. It is the standard chord construction, not a novelty claim. Fewer variables can produce harder remainder equations; no speed or degree improvement follows automatically.

4. Certified orbit-component removal. On $j = 0$, horizontal lines select full order-three automorphism orbits. When labels are propagated by an eigenvalue λ satisfying $1 + \lambda + \lambda^2 = 0$, both label sums vanish. After checking orbit closure, point labels, sign changes, and eigenvalue action on both G and Q, the builder adds $tz - 1 = 0$ to exclude those components. Independently labeled points on the same geometric orbit can be informative and are refused by this pruning builder. The ordinary $B = 0$ guard remains in recovery. The inverse variable can increase elimination cost; pruning is not an automatic speed improvement.



Verification and limits. The final command receipts and requirement status are in the study README and `validation/`. Seven core-only Lean lemmas cover coefficient expansion, root soundness, sign invariance, orbit-label cancellation and conditional saturation. Their axioms are standard. The full squarefree divisor/chord-law completeness theorem, subgroup sign argument, and Rust correspondence remain a partial formalization; see `PROOF_MAP.md`.

The full release suite passed 5,002 tests, with zero failures and 122 existing ignored tests. Focused checks passed 20 APS, 7 CLI, 63 F4-related and 15 GLV tests. They compare exact relation sets, verify known public answers and test refusal/timeout cases. They share the native curve arithmetic; independent EC replay remains open. The shipped native identity certificate is `IDC1h0abc57c4d1e81eb0`; the example checks its statement fingerprint and evaluates a second seeded set of points. See `line-identity.json`.

Performance remains open. No end-to-end cost, external solver comparison, speedup or scaling is measured. Complete accounting/replay/export and a frozen controlled protocol are still owed in the existing complete-cost panel. Automatic approval review rejected the mandatory resume checkpoint upload because it would package source and local changes for S3; no performance experiment was launched or checkpoint gate bypassed. Local correctness checks and publication of the requested PR proceed separately.

Sources. Amadori–Pintore–Sala (2018), revised manuscript, sections 3.1–3.4; Milne, Elliptic Curves, second edition, I section 3; `research/factor_base_designs_20261015/README.md`, design 11 (including its cost-unfavourable measured cases); `src/cryptanalysis/glv_invariant_base.rs`, existing root extraction. The earlier source map records the Kudo follow-up full-text access gap; no reproduction or novelty claim depends on that inaccessible body.

The IC leaderboard, performance charts and canonical curve graphs receive no new values: this study establishes no qualifying measured gain or new curve route. The editable diagram above records the implemented paths.